

# Most imp AI ML interview question

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## LLM / Generative AI Interview Questions

1. What is an LLM?
2. Difference between AI, ML, Deep Learning, and Generative AI?
3. What is a Transformer architecture?
4. What are tokens?
5. What is tokenization?
6. Explain attention mechanism.
7. What is self-attention?
8. What is positional encoding?
9. Difference between encoder and decoder?
10. What is temperature in LLMs?
11. What is top-k and top-p sampling?
12. What is hallucination in LLMs?
13. Why do hallucinations happen?
14. How do you reduce hallucinations?
15. How do you handle hallucination?
16. What is prompt engineering?
17. What is context window?
18. What is fine-tuning?
19. Difference between fine-tuning and prompt engineering?
20. What is RAG (Retrieval-Augmented Generation)?
21. Why use vector databases?

## Production / Real-World AI Questions

22. Why does your model work in development but fail in production?
23. Why does your model work in dev but not in production?
24. How do you monitor ML models in production?
25. What is model drift?
26. What is data drift?
27. How do you reduce LLM cost?
28. How do you reduce LLM Cost?
29. How do you optimize latency in AI systems?
30. How do you scale an LLM application?
31. How do you cache LLM responses?
32. How do you evaluate LLM performance?
33. How do you handle rate limits from APIs?
34. What happens if the API goes down?
35. How do you improve inference speed?
36. What is quantization?
37. What is batching?
38. How do you secure an AI application?

## RAG Interview Questions

39. Explain the RAG pipeline end-to-end.
40. What are embeddings?
41. Which embedding models have you used?
42. What is semantic search?
43. Difference between keyword search and vector search?

- 44. What is chunking?
- 45. Best chunk size for documents?
- 46. What vector databases have you used?
- 47. How do you improve retrieval quality?
- 48. What is reranking?
- 49. What is hybrid search?
- 50. How do you avoid irrelevant context retrieval?
- 51. How do you evaluate a RAG system?

## Prompt Engineering Questions

- 52. What makes a good prompt?
- 53. Explain zero-shot, one-shot, and few-shot prompting.
- 54. What is chain-of-thought prompting?
- 55. What is role prompting?
- 56. How do you structure prompts for reliability?
- 57. How do you reduce hallucination using prompts?
- 58. What is prompt injection?
- 59. How do you defend against prompt injection attacks?
- 60. What is system prompt vs user prompt?
- 61. How do you test prompts?

## Deep Learning Questions

- 62. What is backpropagation?
- 63. What is gradient descent?
- 64. Difference between CNN, RNN, and Transformer?
- 65. What is overfitting?

- 66. What is dropout?
- 67. What are activation functions?
- 68. What is vanishing gradient?
- 69. Explain batch normalization.
- 70. What optimizers have you used?
- 71. Difference between Adam and SGD?

## Python + ML Engineering Questions

- 72. Difference between list and tuple?
- 73. What are generators in Python?
- 74. What is multithreading vs multiprocessing?
- 75. Explain decorators.
- 76. How does memory management work in Python?
- 77. Explain NumPy broadcasting.
- 78. Difference between Pandas loc and iloc?
- 79. What is vectorization?
- 80. Why is NumPy faster than Python loops?
- 81. Explain OOP concepts in Python.

## Data Science / ML Questions

- 82. Difference between supervised and unsupervised learning?
- 83. Bias vs variance?
- 84. Explain cross-validation.
- 85. What is feature engineering?
- 86. What is regularization?
- 87. Difference between L1 and L2?

- 88. Explain precision, recall, and F1-score.
- 89. What is ROC-AUC?
- 90. How do you handle imbalanced datasets?
- 91. Explain confusion matrix.

## System Design for AI Interviews

- 92. Design a ChatGPT-like application.
- 93. Design a document Q&A; system using RAG.
- 94. Design an AI chatbot for customer support.
- 95. How would you serve millions of AI requests?
- 96. How do you reduce AI inference cost?
- 97. How would you build a resume screening system?
- 98. Design a recommendation system.
- 99. How would you deploy an ML model?
- 100. Explain CI/CD for ML systems.
- 101. What is MLOps?

## Advanced LLM Questions

- 102. What is LoRA?
- 103. What is QLoRA?
- 104. What is RLHF?
- 105. What are agents in AI?
- 106. What is function calling?
- 107. MCP vs API?
- 108. What are tool-calling agents?
- 109. What is memory in AI agents?

- 110. Explain Mixture of Experts (MoE).
- 111. What is speculative decoding?
- 112. What is distillation?
- 113. What are open-source LLMs you know?

## Frequently Asked HR + Scenario Questions

- 114. Tell me about an AI project you built.
- 115. Biggest challenge you faced in ML?
- 116. How do you debug a poor-performing model?
- 117. Describe a production issue you solved.
- 118. Explain a complex AI concept to a non-technical person.
- 119. What would you do if users complain about wrong AI answers?
- 120. How do you prioritize speed vs accuracy?
- 121. Have you worked with cloud platforms?
- 122. Have you deployed models?
- 123. Why should we hire you for this AI role?